

Winsome Resources

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Winsome Resources

Company or organization location (we welcome applicants from anywhere in the world)

Québec, Canada

Name(s) of primary point(s) of contact for this application

Decarbonization Manager, Winsome Resources, Gary Stewart

Brief company or organization description <20 words

Winsome Resources is an Australian lithium developer company that is developing the world's first mining CDR hub

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

Winsome Resources is transforming the former Renard diamond operation in Québec, Canada into a world leading Carbon Dioxide Removal (CDR) hub. The repurposing of the operations extensive diamond mining by-products (legacy processed kimberlite material and waste ultramafic rock) and existing processing infrastructure will unlock a pathway for permanent CO₂ removal at industrial scale. The first-of-its-kind mining-based CDR hub will allow multiple mineralisation technologies to be piloted, scaled and deployed.

The former Renard diamond operation has extensive existing infrastructure to support the proposed CDR Hub including an airport, power station, processing plant, water treatment plant, workshops, processed mineralised material storage and a substantial camp.

Winsome will leverage multiple accelerated mineral carbonation processes at the proposed Renard CDR Hub. Mineral Carbonation, also commonly known as Carbon Mineralisation, is a natural process where metal cations (such as Mg²⁺ and Ca²⁺), released via the weathering and dissolution of suitable mineral groups, bond with CO₂ to form carbonate minerals – which can be durably stored on geological timescales (>1000 years).

The Renard CDR project will involve collaboration and partnerships with multiple Mineralisation based CDR suppliers. Partnered suppliers will work with Winsome to utilize the Renard operations >13 Mt of magnesium-rich processed kimberlite material, which contains suitable minerals such as *brucite*, *serpentine*, *olivine*, as a feedstock for various mineral carbonation technologies. The Renard CDR hub will serve as a first-of-its-kind model for turning legacy mining operations into climate-positive assets, providing a scalable template for gigatonne-scale CO₂ removal globally.

The CDR systems deployed at Renard will harnesses and accelerate natural mineralization processes, durably storing atmospheric CO₂ over geologic timescales. Mineral Carbonation methods may differ by deployed technology, but commonly include the following geochemical processes:

- CO₂ Conversion: Carbon dioxide from the atmosphere (or a concentrated CO₂ stream) is injected and dissolved into the water, where it forms carbonic acid, as well as other dissolved carbon species (such as bicarbonate).
- Mineral Dissolution: Alkaline mineral feedstocks, containing Mg and Ca rich silicate minerals (for example, *brucite*, *serpentine*, *olivine*), react with CO₂-charged water (containing carbonic acid), which causes mineral dissolution and subsequent release of Mg²⁺ and Ca²⁺ ions.
- Carbonate Precipitation: Free Mg²⁺ and Ca²⁺ ions bond with bicarbonate in solution producing carbonate mineral phases – principally magnesium carbonate (MgCO₃, *magnesite*) and calcium carbonate (CaCO₃, *calcite*). These carbonate phases may precipitate in solid form under an alkaline pH (>8.3), storing carbon in a stable mineral form under standard atmospheric conditions at the earth's surface.

The outcome of these processes is durable CO₂ storage – stored carbon is fixed in mineral form that will remain stable for millennia, far exceeding 1,000-year durability. Depending on the individual mineralisation process employed at the Renard Project, it is proposed that produced carbonate minerals will be stored within the existing waste facilities, or be utilised as in materials (such as concrete).

In this project's initial phase, we are rigorously analysing the kimberlite processed material stockpiled at the Renard Operation to quantify the abundance of key reactive minerals (notably *brucite*, *olivine* and *serpentine*). This comprehensive feedstock characterization (through drilling,

sampling, and lab analysis) is crucial to determine the site's exact carbon capture potential and to guide our choice of mineralization techniques.

Armed with this data, Winsome's strategy is to deploy optimal technologies tailored to the mineralogy of the material at Renard. Rather than committing to a single predetermined technology, we take a technology-agnostic approach; we will match the site's mineral profile with the best-suited CO₂ capture and removal methods from our network of partners. By doing so, we ensure maximum efficiency and CO₂ removal yield, rather than forcing a one-size-fits-all solution onto a potentially suboptimal feedstock. We have assembled a group of specialized partners contributing expertise in various accelerated mineralization techniques, including thermal/chemical activation and aqueous approaches, alongside leading researchers. This collaborative, *mineralogy-first* deployment model is a key differentiator for the Renard CDR hub.

Why Our System Is Best-in-Class and Differentiated?

We believe the Renard CDR hub represents a best-in-class carbon removal project, distinguished by several unique strengths:

- **Leveraging an Existing Mine Site & Infrastructure:** Our project uniquely repurposes a former mine and its substantial infrastructure for CDR. The Renard Operation (currently under care & maintenance) already has power supply, roads, water management, IT networks, buildings, and a local workforce in place. Utilizing this existing industrial asset base confers significant advantages: it lowers upfront capital needs and operating costs compared to a greenfield project and avoids the environmental disturbance of building an all-new facility. Moreover, by using remnant material stored on-site following processing as our feedstock, we turn an environmental liability (waste rock) into a valuable resource for CO₂ removal, eliminating the need to extract or transport new minerals. This reuse of a legacy site is a novel approach that not only cuts costs and speeds up deployment, but also exemplifies sustainable innovation in the mining sector.
- **Flexible, Optimized Technology Selection:** Unlike organizations pursuing a single CO₂ removal technique, Winsome employs a technology-agnostic, mineralogy-driven strategy. We will select and combine the best available CDR technologies based on the specific mineral composition identified at Renard. This flexible model ensures we always use the most effective carbonation pathway for the given material – whether that is thermal activation, chemical extraction, direct air CO₂ capture, or other enhancement methods. By optimizing technology to fit the feedstock (rather than the other way around), we maximize carbon removal efficiency, de-risk the operation, and maintain adaptability as new mineralization innovations emerge. This approach differentiates us from similar projects that are often locked into a single technology or methodology regardless of site specifics.
- **Collaboration with Leading Experts:** Rather than developing a single proprietary technology in isolation, we are partnering with a diverse group of CDR technology providers and researchers to build the Renard hub. We are bringing together top innovators in the field – for example, providers of microwave-assisted activation, chemical leaching methods, pit lake alkalinity enhancement, and leading university researchers – each contributing cutting-edge solutions for different aspects of mineral carbonation and CO₂ capture. By leveraging this broad expertise, we can combine and integrate multiple innovations into one project, something a single-technology company cannot easily do. Our role is to coordinate and optimize conditions for piloting these technologies at the Renard site. This collaborative model accelerates development (by using proven methods from each partner), reduces technical risk (through diversification), and strengthens our overall solution, making it truly state-of-the-art. No other organization pursuing mineralization is operating as a hub of

multiple technologies in this way.

- Potential for Dual-Use (Mineral Recovery + CDR):** Winsome is also exploring a dual-use future for the Renard site that further differentiates our approach. In parallel with carbon removal, the site may host commercial mineral extraction – specifically, leveraging the carbon mineralization process alongside potential lithium recovery from adjacent ore bodies (such as Winsome’s nearby Adina project). If successful, this would create a climate-positive mining operation where critical minerals (rich in battery metals such as Cs, Ta and Li) are produced while simultaneously mineralizing CO₂ from the atmosphere. This vision of integrating resource production with CO₂ sequestration is highly innovative. Such an approach would enhance the project’s economics and demonstrate how carbon removal can be embedded into the mining industry’s core business, turning an operation into a net-positive asset from both a commodity production and climate perspective. This concept is an emerging opportunity unique to our project’s context and partnerships.
- Independent Validation & Rigor:** We have prioritized a science-driven and transparent approach from the start. Winsome has commissioned an independent carbon mineralization study by WSP Canada, a leading engineering firm, to validate the CO₂ storage potential at Renard. WSP’s analysis is quantitatively assessing the mineral carbonation capacity of the stockpiled processed material and waste rock and developing a roadmap for scaling up the process. This third-party validation provides an objective check on our assumptions and lends credibility to our proposal, ensuring that our plans are grounded in solid data and realistic projections. Furthermore, we are committed to rigorous monitoring, reporting, and verification (MRV) of all CO₂ removed. We are collaborating with Isometric (a specialized carbon registry) to implement a comprehensive MRV framework aligned with emerging standards for ex-situ mineralization. This means we will meticulously track CO₂ inputs and outputs, reaction extents, and long-term storage integrity, with transparent reporting and third-party verification of the results. By investing early in independent studies and robust MRV, we differentiate our project as one that meets the highest standards of scientific integrity and creditworthiness.

Alignment with Frontier’s Priority Innovation Areas

Our project design deliberately addresses multiple *Frontier Priority* areas highlighted in the RFP, demonstrating strong alignment with Frontier’s innovation objectives:

Leveraging Existing Assets – *Using and repurposing existing industrial infrastructure.*

We are converting the Renard operation’s existing site infrastructure and equipment into a CDR facility, rather than building from scratch. By tapping into an idle mine’s power systems, roads, and accommodations, we minimize new construction and reduce the environmental footprint and cost of our operations. This exemplifies how legacy industrial sites can be turned into climate solutions, extracting new value from prior investments.

Mine Waste Utilization – *Utilizing kimberlite processed material or waste for carbon removal.*

The core feedstock for our CO₂ removal process is the abundant ultramafic mine waste on-site (millions of tonnes of processed kimberlite processed material and waste rock). We directly convert this waste material into a carbon removal feedstock, which not only avoids waste remediation issues but actively contributes to climate mitigation. In doing so, we transform a potential long-term environmental liability (mine waste storage) into a valuable asset for carbon removal. This approach showcases an innovative form of mining waste upcycling that could be replicated at other mine operations worldwide.

Durable Carbon Storage – Ensuring >1,000-year CO₂ storage longevity.

Mineral carbonation inherently provides durable CO₂ storage in the form of solid carbonate. Once CO₂ is mineralized as carbonate, it will not revert back to gas, even over geological timescales under normal environmental conditions. This permanence far exceeds 1,000 years, meeting Frontier's highest durability criteria. Additionally, because the storage medium is stable carbonate, that will remain on site or be utilised as built materials post carbonation, there is reduced reversal risks. Our approach therefore offers one of the most reliable, low-risk long-term storage solutions available. It is also highly scalable – the Renard site alone has a vast quantity of reactive minerals, and many other mine sites across the globe collectively present gigatonne-scale opportunities for durable CO₂ storage (especially those which mined ultramafic rocks).

Comprehensive MRV – Innovations in measurement, reporting, and verification.

We are building a robust MRV framework from day one, collaborating with CDR partners and Isometric to identify suitable Protocols for the technologies we plan to employ. Every ton of CO₂ removed and mineralized will be carefully measured and verified independently. For example, we will monitor CO₂ capture rates in the field (via gas flow meters, concentration sensors, and periodic sampling of carbonates), and cross-verify carbon uptake with geochemical analyses of the reacted material. Data will be logged and reported in compliance with a registry standard and relevant Protocol, enabling third-party VVB's to certify our removal claims with confidence. By adopting transparent and high-precision MRV practices, we ensure the credibility of our carbon removals and help advance industry best-practices for measuring CO₂ mineralization in open and closed systems.

In summary, the Renard CDR hub will durably remove CO₂ from the atmosphere by turning mine waste into rock-solid carbon storage, leveraging a unique combination of site resources, innovative technologies, and partnerships. This project is designed to be scalable, verifiable, and highly durable, setting it apart as a best-in-class CDR solution. The support from Frontier's prepurchase will enable us to complete the critical feedstock characterization and initial pilot deployments, catalysing the transition of Renard from a dormant mine into a world-leading carbon removal facility. With its technology-agnostic approach and alignment with key innovation priorities, Winsome's Renard CDR hub is positioned to deliver exceptional, best-in-class CO₂ removal outcomes and to serve as a strategic blueprint for climate-focused transformation of industrial sites globally. The presence of existing infrastructure.

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

The key objective of this project is to establish the Renard Operation as a dedicated Carbon Dioxide Removal (CDR) hub. Located in the Eeyou Istchee James Bay region of Québec, Canada, Renard has existing infrastructure and abundant reactive mineral feedstocks that are suitable for multiple

² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

CDR approaches. This first of kind project will provide a testing ground for the large-scale deployment of CDR approaches on mining operations globally. The proposed CDR approach centres on accelerated mineral carbonation, where CO₂ reacts with alkaline materials to form stable carbonate minerals for durable long-term storage.

Winsome aims to shift towards a dual-purpose climate-positive operation, integrating carbon mineralization with the potential for future lithium extraction. The project ultimately aims to galvanize gigatonne-scale carbon removal by providing a scalable model for mine sites globally. The initial focus will be on characterizing the on-site mineral feedstocks and assessing suitable pilot technologies for deployment.

The current levelized cost for the initial pilot at the Renard mine is in the high triple-digit USD per tonne range. This is due to upfront investments for feedstock quantification and pilot technologies, as well as operational expenses, including labour, energy, and MRV development/implementation. Winsome acknowledges that initial costs may differ from full scale operational costs, due to the operation being under Care & Maintenance (C&M) at present. The current Frontier application seeks funding primarily for the purpose of further feedstock characterization through drilling and sampling, to complement the assessments undertaken by WSP.

To approach Frontier's cost target of below \$100 per tonne and achieve gigatonne-scale carbon removal, Winsome intends to pursue several strategies:

- **Aggressive Scaling:** Expand the volume of feedstock utilised for different CDR approaches as the hub develops in scale.
- **Process Optimization:** Refine deployed carbon mineralization processes based on detailed mineralogical characterization, in order to enhance efficiency and reduce costs. This involves selecting the best-suited technologies based on the specific mineral and elemental compositions.
- **CO₂ Capture Integration:** Explore the potential for on-site CO₂ capture. Although surficial mineralisation approaches will initially be prioritised, Winsome aims to explore and possibly integrate Direct air Capture (DAC) on site, where economically and operationally feasible.
- **Formalized Mineral Resource Assessments:** Conduct comprehensive drilling and sampling programs to accurately quantify reactive mineral feedstocks (See Figure 1). This is a primary objective of the current funding request. WSP has been commissioned for a study to assess the preliminary carbon mineralization potential, based on pre-existing elemental data.
- **Strategic Financial Leveraging:** Utilize financial incentives like potential tax credits (Such as SR&ED, CCUS ITC, Quebec C3i). Winsome plans to explore national tax credit schemes, related both to decarbonisation and critical mineral production within Canada. Such schemes will allow Winsome to reduce gross tonnage costs over the project lifetime.
- **Leveraging Existing Mine Infrastructure and Workforce:** Capitalize on the existing infrastructure at the Renard Operation (including energy, IT, site services and transportation) under C&M and the availability of a skilled workforce to potentially reduce costs. Existing roles at the operation can be aligned with potential CDR roles.
- **Sustainable Energy Integration:** A core objective is to power the Renard CDR Hub using low-carbon energy. Our roadmap includes assessing the existing energy infrastructure's capacity, evaluating the potential for integrating on-site renewable energy generation (such as solar or wind, leveraging the site's footprint), and exploring Power Purchase Agreements

(PPAs) for certified renewable energy. This phased transition towards sustainable power is crucial for minimizing the operational carbon footprint, ensuring the overall carbon negativity of the process, and contributing to the project's long-term cost-effectiveness and environmental integrity.

- **Collaboration and Technology Selection:** Strategically select and optimize technologies from CDR partners, including promising technology like Arca Climate (microwave activation), Exterra Carbon Solutions (mineralization and low-carbon alkalinity extraction), and Aquarry Earth (pit lake based CO₂ capture and mineralization). Winsome adopts a technology-agnostic approach based on the specific mineralogy of feedstocks at Renard, Winsome aims to make this project a testing bed for multiple suppliers who have the viability to achieve industrial scale CDR.

Removal activities undertaken at the Renard operation will be quantified and verified in line with Isometrics Registry Standard, Pathway level protocols and storage modules. Where Protocols do not currently exist for potentially implementable technologies, Winsome will work with Isometric to identify the appropriate MRV quantification approach. At a high level MRV approaches will involve:

- **Baseline scenario and counterfactual weathering estimations** – This will include distinct control plots and an assessment of pre-existing feedstocks on site prior to CDR intervention
- **Project Additionality** – All project activities will be designed to demonstrate financial, environmental and regulatory additionality, in line with ICVCM core principles
- **Periodic Sampling and Analysis** - Regularly collecting samples (solid, aqueous and gaseous phase) in project control and deployment areas to ensure accurate quantification of removals and additionality.
- **Continuous Monitoring** - Ongoing measurements will be collected through the projects reporting periods (RP) to verify the stability and durability of stored captured atmospheric CO₂. Continuous monitoring will be carried out for the purpose of removal quantification and environmental protection, ensuring technologies do not result in net environmental harm.
- **Third-Party Verification** - To ensure accuracy in removals, and reduce the risk of perceived greenwashing, all removal activities will be reported in a Project Design Document (PDD). PDD's will be designed to meet the requirements of Isometric Protocols, with independent 3rd party verification on site by Validation and Verification Bodies (VVB's)

Detailed MRV plans will be identified in line with partnered CDR suppliers upon commencement of the Renard Hub. As Renard will be a technology agnostic CDR hub, multiple MRV approaches will be required. At this stage Winsome plans to employ Protocols, such as Isometrics Open System Ex-situ Mineralization (OSEM), to quantify surficial approaches employed by certain partners.

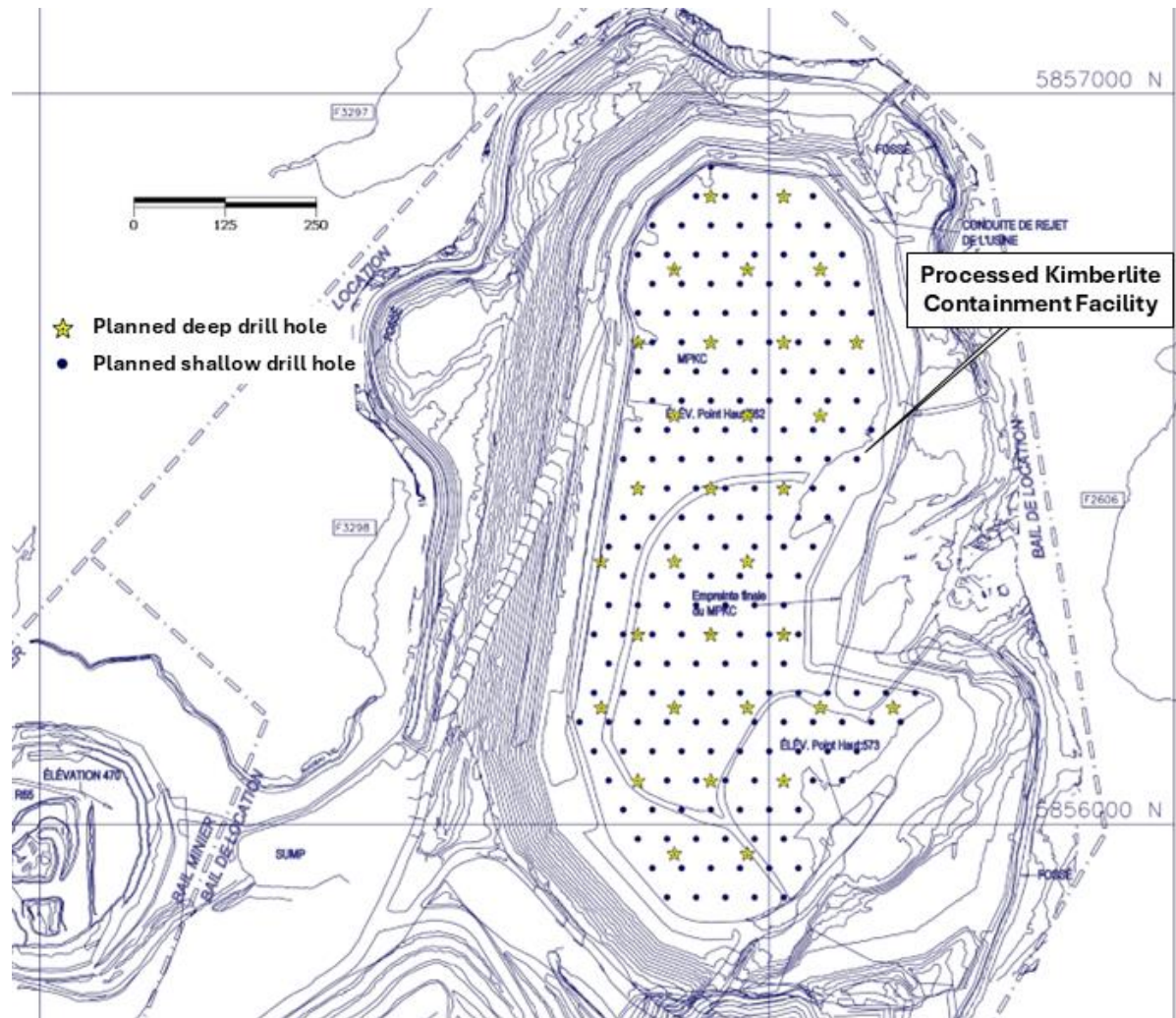
The current Frontier funding application is strategically focused on the crucial first step of detailed feedstock characterization. The resulting data will guide the determination of how best to allocate feedstock materials across removal projects. Winsome proposes an initial commitment of 400 net tonnes of CO₂ removal to Frontier over a 3-year period. Based on existing market interest demonstrated by technology partners. These tonnes will be delivered across multiple pilot deployments following feedstock characterization at Renard. This volume is subject to refinement following the proposed drilling and characterisation program.

Funding from Frontier will be specifically directed towards critical early-stage activities, including:

- **Drilling and sampling programs:** These are necessary to extract samples for analysis

from the kimberlite processed material (see Figure 1) and waste rock at various depths and locations across the site. This will provide the raw material for detailed analysis.

Figure 1. Processed. Kimberlite Containment Facility.



- **Comprehensive mineralogical and elemental studies:** Geochemical, mineralogical and elemental analysis will be undertaken, using techniques such as SEM-EDX, XRD, ED-XRF and Total Carbon (TC), on samples collected from existing storage facilities at Renard to quantify mineral abundances, potential reactivity and carbonation potential. This includes detailed mapping and kimberlite processed material characterization.
- **Detailed infrastructure assessments:** The funding will support evaluations of the existing infrastructure at the Renard site to determine its suitability for accommodating multiple simultaneous CDR pilot technologies from Winsome's partners. This includes assessing energy requirements, water access, and transportation networks.
- **Permitting studies:** Securing regulatory approvals is essential for the deployment of onsite

CDR pilots. Frontier funding will contribute to the necessary studies and consultations to facilitate these approvals.

The initial funding phase will focus on feedstock characterization, securing necessary permits and assessing overall suitability of planned pilot deployments. The information obtained from these activities will be instrumental in guiding the selection and deployment of the most appropriate and cost-effective CDR technologies at Renard. Without targeted Frontier funding, Winsome will have difficulty securing alternative funding specifically to assess carbon removal potential and to pave the way for on-site piloting. This funding is therefore a critical catalyst for developing the world's first CDR hub at an existing mining operation.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Importance of the WSP Study for Technology Validation

The WSP Carbon Mineralization Potential Study is critical in validating historical mineralogical data at the Renard site, specifically by confirming the concentration, distribution, and reactivity of key minerals within the kimberlite processed material and waste rock. This validation will directly inform which carbon dioxide removal (CDR) technologies could be effectively employed at the Renard location. By precisely characterizing the mineralogical variability through detailed drilling, sampling, and laboratory analysis, Winsome will reduce uncertainties and provide site-specific insights into the feasibility, economic viability, and potential scalability of various mineral carbonation technologies. The data derived from the WSP study will thus enable Winsome and potentially other CDR technology providers to assess compatibility with Renard's unique conditions, improving market certainty and promoting industry-wide learning.

Technical Risks - Uncertainty in Mineral Carbonation Potential: There's a risk that the actual rate, extent, and effectiveness of mineral carbonation under real-world conditions at Renard may differ from predictions due to variability in mineralogy, particle size, moisture content, temperature, and accessibility of reactive minerals.

Mitigation: Winsome will conduct a detailed mineralogical characterization of the feedstock through extensive drilling, sampling, and laboratory analysis, including the ongoing WSP Carbon Mineralization Potential Study, to validate historical data and provide site-specific insights. This will reduce uncertainties and inform the feasibility and scalability of various carbonation technologies.

Performance and Reliability of Pilot Technologies: The Performance and Reliability of Pilot Technologies: The performance and reliability of various emerging CDR pilot technologies (e.g., those involving **microwave activation**, chemical leaching, and pit lake enhancement) under the specific conditions at the Renard site are uncertain, as they have varying Technology Readiness Levels (TRLs) and have not yet been validated on-site."

Mitigation: Winsome will conduct detailed, phased testing programs to characterize site conditions before full-scale pilot deployment. Early-stage studies, including mineralogical characterization and infrastructure assessments, will provide crucial baseline data to

validate the suitability of specific technologies. This approach maximizes adaptability and allows technology providers to assess their compatibility with the Renard site.

Site Readiness for Pilot Technologies: There's a technical risk related to whether the existing infrastructure (power, water management, logistics, accessibility) is suitable for hosting multiple potential CDR pilot technologies.

Mitigation: Winsome will conduct an extensive infrastructure review to assess and confirm the site's readiness to support scalable pilot projects.

Project Execution Risks:

- **Funding Availability for Early-Stage Development and Testing:** Securing upfront funding for essential initial characterization studies (drilling, sampling, mineralogical assessments) is a significant risk, as traditional mining companies may not prioritize such early-stage CDR feasibility studies.
 - **Mitigation:** Winsome is seeking Frontier funding specifically for these early-stage activities, emphasizing that this funding is a critical catalyst. A pre-purchase agreement with Frontier for an initial amount of carbon removal will also provide crucial early-stage funding.
- **Permitting and Regulatory Approvals:** Obtaining the necessary permits and regulatory approvals from federal (NRCan) and provincial (Quebec) authorities for large-scale carbon mineralization initiatives is a critical risk, given the evolving regulatory landscape in Canada.
 - **Mitigation:** Winsome will proactively engage with regulatory authorities early in the project development cycle to clarify permitting processes, identify requirements, and align project objectives with government priorities. Early preparation and submission of detailed project plans will facilitate timely review.

Measurement, Reporting, and Verification (MRV) Risks:

- **Accuracy and Reliability of CO₂ Removal Quantification:** Accurately quantifying the CDR achieved at the Renard site is technically challenging due to variability in feedstock, environmental factors, and measurement techniques, which could impact the credibility of carbon credits.
 - **Mitigation:** Winsome is partnering with Isometric to establish a MRV infrastructure. We will work with Isometric to apply appropriate uncertainty discounts and buffer pools to CDR figures to ensure reported net removal is conservative and credible. Comprehensive baselining, continuous ecological, environmental and storage monitoring will all be implemented across the project RP and post closure. All removal activities will be designed in line with ICVCM requirements.

Ecosystem Risks:

- While no negative impacts on riverine, coastal, or ocean ecosystems are anticipated due to existing wastewater treatment systems in place at Renard, unforeseen ecological impacts may occur as a result of those pilots which involve water bodies.
 - **Mitigation:** Winsome, in collaboration with partners, will implement comprehensive baseline and continuous ecological monitoring, including regular assessments of water chemistry, sediment composition, and benthic ecosystem health, to swiftly identify and address any unforeseen impacts.

Financial Risks:

- **Cost Overruns and Funding Gaps:** Establishing and operating a scalable CDR hub will

require significant capital and ongoing expenditures, with potential risks of cost overruns and funding gaps beyond the initial pilot phase.

- **Mitigation:** Winsome will develop a detailed project budget with contingency plans. Securing a prepurchase agreement with Frontier will provide critical early funding. Demonstrating the success and cost-effectiveness of initial pilots, along with the validation of industrial scale CDR within a mining operation, will enhance the project's attractiveness to future investors and CDR buyers.
- **Force Majeure Events:** Unanticipated events like extreme weather or natural disasters could disrupt project timelines.
 - **Mitigation:** Winsome will develop comprehensive contingency plans and maintain adequate insurance coverage.

Winsome's overall strategy emphasizes a phased approach, starting with thorough feedstock characterization to reduce technical and financial uncertainties before committing to specific pilot technologies and large-scale deployment. The Frontier funding is crucial to overcome the initial hurdle of funding this essential characterization phase.

d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	Winsome Resources is proposing an initial commitment of 400 net tonnes of CO ₂ removal over a 3-year period.
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	Delivery window: The proposed delivery window for this initial 400 tonnes would be over three to four years following the deployment of initial pilot technologies, potentially commencing in late 2027 or early 2028. This timeline allows for the necessary drilling, sampling, data analysis, and pilot project setup.
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	Levelized cost (\$/ton CO₂): At this early stage, with the primary focus on feedstock characterization and suitability assessments, it is premature to provide a precise levelized cost per tonne for future CDR deployment. The eventual cost will be heavily

	dependent on the specific CDR technologies chosen based on the mineralogy results and their scalability at the Renard site operation. Winsome aims to be make Renard a first of a kind testing ground for mineralisation and mining based CDR technologies. As highlighted previously, initial costs are anticipated to be higher due to the nature of being a mining operation under Care & Maintenance (C&M) rather than a new startup. The funding from Frontier is critical to enabling the feedstock characterization and initial pilot deployment, which will then inform the most cost-effective CDR strategies.
Levelized price (\$/ton CO₂)³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	Winsome Resources understands Frontier's interest in cost-effective CDR solutions. While the long-term target will be to align with Frontier's criteria, for this initial commitment tied to the funding for feedstock characterization and initial deployment development, we are proposing a levelized price of \$3000 per net tonne of CO₂ removed. This price reflects the current early-stage nature of the project, the costs associated with establishing a CDR hub at a remote mine site, and the value of undertaking rigorously verified and durable carbon removal. It is crucial to reiterate that the primary objective of this application is to secure funding for the essential first step of feedstock characterization and initial deployment development. The data obtained will be instrumental in determining the most suitable and scalable CDR technologies for the Renard site and in providing more accurate projections for future CDR volume, cost, and price. This initial commitment of 400 tonnes serves as a tangible demonstration of Winsome's intent and

³ This does not need to exactly match the cost calculated for “This Project” in the TEA spreadsheet (e.g., it’s expected to include a margin and reflect reductions from co-product revenue if applicable).

	the potential of the Renard site as a industry scale CDR hub.
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